Daily Logic Drill #2

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Implication, Converse, Inverse & Contrapositive · Necessary vs Sufficient.

Section A Rapid Recognition

1. For real numbers x, y, classify "$x > y$" as a condition for "$x^3 > y^3$".

Ans: Necessary and sufficient.

2. For non-zero real numbers x, y, classify "$x > y$" as a condition for "$\frac{1}{x} < \frac{1}{y}$".

Ans: Neither necessary nor sufficient.

3. For the quadratic equation $x^2 - 2kx + 1 = 0$, classify "$k > 1$" as a condition for "the equation has two distinct positive real roots".

Ans: Necessary and sufficient.

4. For real numbers x, y, classify "$|x + y| = |x| + |y|$" as a condition for "$xy \geq 0$".

Ans: Necessary and sufficient.

5. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. Classify "$f'(x) > 0$ for all $x \in \mathbb{R}$" as a condition for "f is strictly increasing on $\mathbb{R}$".

Ans: Sufficient but not necessary.

6. For real constants a, b, c, d, classify "$ad - bc \neq 0$" as a condition for "the system $ax + by = 0,\ cx + dy = 0$ has only the trivial solution $(x, y) = (0, 0)$".

Ans: Necessary and sufficient.

7. For a point $(x, y) \in \mathbb{R}^2$, classify "$x^2 + y^2 \leq 1$" as a condition for "$|x| + |y| \leq \sqrt{2}$".

Ans: Sufficient but not necessary.

8. For a positive integer n, classify "$12 \mid n$" as a condition for "$4 \mid n$ and $6 \mid n$".

Ans: Necessary and sufficient.

9. For a triangle with side lengths a, b, c, classify "$a^2 + b^2 = c^2$" as a condition for "the area of the triangle is $\frac{1}{2}ab$".

Ans: Necessary and sufficient.

10. For an integer n, classify "n is odd" as a condition for "$n^2 \equiv 1 \pmod 8$".

Ans: Necessary and sufficient.

Section B Manipulation Drills

1. Determine the necessary and sufficient condition on real k such that the quadratic equation $x^2 - kx + 9 = 0$ has two distinct real roots that are both strictly greater than 1. *(after TMUA 2021 Paper 2 Q9)*

Ans: $6 < k < 10$.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. Consider the conditions:

I. $f(-x) = f(x)$ for all $x \in \mathbb{R}$.

II. $f'(-x) = -f'(x)$ for all $x \in \mathbb{R}$.

Classify condition I as a condition for condition II. *(after TMUA 2018 Paper 2 Q10)*

Ans: Necessary and sufficient.

3. For positive real numbers x, y, classify "$\ln(x + y) < \ln x + \ln y$" as a condition for "$x > 1$ and $y > 1$".

Ans: Neither necessary nor sufficient.

4. State the contrapositive of: "If $a^2 + b^2$ is divisible by 3, then both a and b are divisible by 3" (for integers a, b). Is this contrapositive true?

Ans: Contrapositive: "If at least one of a or b is not divisible by 3, then $a^2 + b^2$ is not divisible by 3." True.

5. Let T_1 and T_2 be two triangles. Classify "T_1 and T_2 have equal perimeter and equal area" as a condition for "T_1 and T_2 are congruent".

Ans: Necessary but not sufficient.

6. For an integer n, classify "n is odd" as a condition for "$n^3 - n$ is a multiple of 24".

Ans: Sufficient but not necessary.

7. Suppose propositions P, Q, R satisfy $P \implies Q$ and $R \implies Q$. Must either $P \implies R$ or $R \implies P$ be true? Justify with a mathematical counterexample.

Ans: No. (Counterexample: let Q be $x^2 \geq 0$, P be $x = 1$, and R be $x = -1$).

8. For real c, determine whether "$c < 0$" is necessary, sufficient, both, or neither for the quartic equation $x^4 + x^2 + c = 0$ to have at least two distinct real roots.

Ans: Necessary and sufficient.

9. Consider the statement: "If n is a multiple of 9, then n^2 is a multiple of 27" (for positive integer n). State its converse, and determine whether the converse is true or false.

Ans: Converse: "If n^2 is a multiple of 27, then n is a multiple of 9." True.

10. Let $f : \mathbb{R} \to \mathbb{R}$ be twice differentiable. Classify "$f'(a) = 0$ and $f''(a) > 0$" as a condition for "f has a local minimum at $x = a$".

Ans: Sufficient but not necessary.

Section C Substitution & Structure

1. For real numbers x, y, the condition "$x + y$ is an integer and $x - y$ is an integer" is:

Ans: A) necessary but not sufficient for "x and y are both integers".

2. Determine the condition on real parameters p, q that is necessary and sufficient for the cubic $x^3 - 3px + q = 0$ with $p > 0$ to have three distinct real roots.

Ans: B) $4p^3 > q^2$.

3. Classify "a triangle has all three altitudes of equal length" as a condition for "the triangle is equilateral". Justify using the fact that an altitude to side a has length $\frac{2\times\text{Area}}{a}$.

Ans: C) necessary and sufficient.

4. For positive integers a, b, n, suppose you're told only that n divides a, or n divides b (possibly both). Does this pin down whether n divides the product ab? And working the other way: does knowing n divides ab pin down whether n divides a or n divides b? *(after TMUA 2019 Paper 2 Q5)*

Ans: B) "n divides a or n divides b" is sufficient but not necessary for "n divides ab".

5. For a twice-differentiable function $f : \mathbb{R} \to \mathbb{R}$, which of the following is necessary and sufficient for the midpoint convexity condition $\frac{f(a)+f(b)}{2} \geq f\left(\frac{a+b}{2}\right)$ to hold for all $a, b \in \mathbb{R}$? *(after TMUA 2018 Paper 2 Q18)*

Ans: B) $f''(x) \geq 0$ for all $x \in \mathbb{R}$.

6. Classify "n^2 is even" as a condition for "n is even".

Ans: C) necessary and sufficient.

7. Let $p(x)$ be a cubic polynomial with real coefficients. Which of the following conditions is necessary and sufficient for the existence of $c \in (a, b)$ such that $p'(c) = 0$? *(after TMUA 2021 Paper 2 Q8)*

Ans: C) $p'(x) = 0$ has real roots and at least one lies in (a, b).

8. State the contrapositive of "If a and b are both irrational, then ab is irrational." Is the original statement true? Give a counterexample if false.

Ans: Contrapositive: "If ab is rational, then a is rational or b is rational." The original statement is false; counterexample $a = b = \sqrt{2}$.

Section D Challenge

1. Let $f(x) = x^2 + bx + c$ with $f(0) = c$. Let P be the statement: "if $c < 0$, then f has two distinct real roots." Which of the following must be true? *(after TMUA 2022 Paper 2 Q5)*

Ans: F) I and III only.

2. Consider a finite list of n real numbers (repeats allowed), sorted into non-decreasing order. Let P be the statement "n is odd", and let Q be the statement "the middle value of the sorted list equals one of the list's own entries". P is: *(after TMUA 2022 Paper 2 Q6)*

Ans: B) sufficient but not necessary for Q.

3. For a positive integer n, let P be "n leaves remainder 1 when divided by 4" and Q be "n^2 leaves remainder 1 when divided by 8." P is:

Ans: B) sufficient but not necessary for Q.

4. Statement S: "If a quadrilateral has four equal sides, then its diagonals are perpendicular." (a) Is S true? (b) State S's converse. Is the converse true?

Ans: (a) True (a rhombus). (b) Converse: "If a quadrilateral's diagonals are perpendicular, then it has four equal sides." The converse is false.

5. Statement S: "If n is a positive integer such that $n^2 - 1$ is prime, then $n = 2$." (a) Is S true? Prove or disprove. (b) State S's contrapositive. (c) Is "$n^2 - 1$ is prime" necessary and sufficient for "$n = 2$"?

Ans: (a) True. (b) "If $n \neq 2$, then $n^2 - 1$ is not prime." (c) Yes.